

Assignment – 12:

Problem Statement:

Write a program in C to accept two numbers from the user and compute:

- a) The smallest common divisor of the two numbers (other than 1, if any)
- b) The Greatest Common Divisor (GCD) of the two numbers

Code:

```
#include <stdio.h>

int main()
{
    int num1, num2, i;
    int smallest_divisor = 0;
    int gcd;
    printf("Enter two numbers: ");
    scanf("%d %d", &num1, &num2);
    int min;
    if (num1 < num2)
    {
        min = num1;
    }
    else if (num2 < num1)
    {
        min = num2;
    }
    for (i = 2; i <= min; i++)
    {
        if (num1 % i == 0 && num2 % i == 0)
        {
            smallest_divisor = i;
```

```

        break;
    }
}
if (smallest_divisor != 0)
{
    printf("Smallest common divisor (other than 1): %d\n", smallest_divisor);
}
else
{
    printf("No common divisor other than 1 exists.\n");
}
int a = num1, b = num2, temp ;
while (b!=0)
{
    temp=b;
    b=a%b;
    a=temp;
}
gcd=a;
printf("GCD of the two numbers: %d\n", gcd);
}

```

Input & Output:

```

Enter two numbers: 55 15
Smallest common divisor (other than 1): 5
GCD of the two numbers: 5

```

```

Enter two numbers: 5 6
No common divisor other than 1 exists.
GCD of the two numbers: 1

```

